Galvanic Cells and Electrode Potentials

Cell potentials, cell notation, ΔG° and electrolysis

Name _____ Class _____ Date _____

STANDARD REDUCTION POTENTIALS (25 °C) AND CONSTANTS

F_2/F^- +2.87 V · Ag^+/Ag +0.80 V · Cu^{2+}/Cu +0.34 V · $2H^+/H_2$ 0.00 V · Ni^{2+}/Ni −0.25 V · Zn^{2+}/Zn −0.76 V · Mg^{2+}/Mg −2.37 V

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ · $\Delta G^\circ = -nFE^\circ_{cell}$ · $Q = It$ · $n(e^-) = Q/F$

F = 96 485 C/mol · 1 C × 1 V = 1 J · 1 A = 1 C/s · Cu = 63.55 g/mol · Ag = 107.87 g/mol

Show units in every step.

A. Understanding

1. In a galvanic cell, state what happens at the anode and the cathode, which way electrons flow through the wire, and the role of the salt bridge.

B. Cell potentials

2. A cell is made from Zn^{2+}/Zn and Ag^+/Ag half-cells. (a) Identify the anode and the cathode. (b) Calculate E°cell. (c) Write the overall equation. (d) Write the cell notation.

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3. Calculate E°cell for a cell made from Ni^{2+}/Ni and Ag^{+}/Ag half-cells, and state which metal is the anode.

4. Calculate E°cell for a magnesium–copper cell.

5. Use E°cell to decide whether each reaction is spontaneous: (a) $Cu + Zn^{2+} \longrightarrow Cu^{2+} + Zn$ (b) $Zn + 2\,H^{+} \longrightarrow Zn^{2+} + H_2$

6. From the data, identify the strongest oxidizing agent and the strongest reducing agent.

C. Free energy

7. Calculate ΔG° for the Daniell cell, $Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$.

D. Electrolysis

8. A spoon is silver-plated using a current of 1.50 A for 45.0 min ($Ag^{+} + e^{-} \longrightarrow Ag$). What mass of silver is deposited?

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E. Challenge

9. How long, in minutes, must a current of 3.00 A flow through copper(II) sulfate solution to deposit 5.00 g of copper ($Cu^{2+} + 2\,e^- \longrightarrow Cu$)?

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